Weighted Distribution Aggregation for Heterogeneous Federated Synthetic Data: A Statistical Framework and Fixed-Target Evaluation

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***Abstract*—Federated tabular synthesis must preserve a declared population target while combining imperfect local generators. We study whether statistical weighting of non-neural client models improves empirical synthetic-data fidelity over locally tuned, schema-corrected sample-size weighting. The server receives fitted models and explicitly declared metadata; privileged real-record access is separated from this release contract. Our framework relates representation constraints, finite-pilot optimization, stratified generation, conditional support, prediction, and release accounting. We give proofs of finite-workload bounds, a sufficient uncertainty-aware fallback certificate, and counterexamples delimiting their interpretation. The certificate is an analytical extension, not an evaluated selection algorithm. In 54 matched UCI Adult development cells, a constrained minimax challenger attained the registered 0.005 improvement in none of 18 three-seed slices; its mean loss difference from size weighting was +0.002428. A written global freeze selected the Gaussian-copula size-weighted procedure. Subsequent official-holdout scoring gave a mean difference of -0.000215 and no practical-gain slice among six. A 12-cell Bank Marketing replication gave -0.001332 and no practical-gain slice among four. Secondary audits expose a large conditional dependency error, unequal-budget downstream utility deficits, and model releases larger than matched raw-table payloads. The main pipeline is nonprivate. The contribution is a reproducible, access-aware comparative negative result and its mathematical boundaries, rather than a universally superior weighting algorithm or a population-level fidelity guarantee.**

***Index Terms*—Federated synthetic data, non-IID tabular data, distribution aggregation, statistical calibration, conditional generation, empirical evaluation, negative results.**

I. Introduction

SUPPOSE several institutions hold tables with the same attributes but different populations. One institution may predominantly observe younger people; another may serve a more highly educated population. They want a joint synthetic table for statistical queries and predictive modelling without transmitting their original records to the synthesis server. A natural construction fits a generator at each institution, releases the generators, and samples from their mixture. This avoids raw-row transmission in the synthesis interface, but does not, by itself, establish privacy, accurate conditional distributions, or a smaller communication payload.

The weighting decision has a specific purpose. If client i represents fraction α_i of the intended target, sampling that client's fitted model with probability α_i is the direct plug-in reconstruction. Alternative weights may compensate for local approximation error. They may also obtain a better aggregate score by suppressing clients whose records are difficult to model. Thus, an apparent improvement can combine genuine error correction, finite-pilot overfitting, and an undesirable change in representation. Keeping the reference target fixed is essential to distinguish these effects.

This study asks: *under a fixed client target and an explicit information contract, does statistical weighting materially improve the empirical fidelity of a synthetic table over a properly tuned size-weighted generator while respecting a representation budget?* The measured objects are existing real reference data D_R and generated data D_S. We do not estimate a distance between two unknown population distributions and relabel it as an observed result.

The project also examines conditional feature generation, income-probability prediction, query diagnostics, finite-pilot optimality, and communication and privacy boundaries. These are related questions, not interchangeable definitions of success. Accurate answers to a finite set of unconditional queries need not imply accurate conditional relationships or downstream predictions. A conditional sampler can obey every requested filter while generating the wrong dependencies inside the selected group.

The empirical answer is negative at the registered practical scale. After correcting structural inconsistencies and matching local tuning, constrained weighting did not establish a 0.005 query-max improvement in the Adult development matrix, the frozen Adult official-holdout comparison, or the external Bank replication. The selected main procedure therefore retains size weighting. We retain favourable individual cells, null ablations, and adverse conditional and utility findings; no post-test method reselection is performed.

Our contributions are threefold. First, we specify a target-aware empirical framework that makes model access, target proportions, synthetic allocation, and evaluation references explicit. Second, we provide a self-contained mathematical treatment of mixture expressivity, representation changes, pilot uncertainty, fallback decisions, projected probabilities, conditional transfer, and bounded-loss utility. These results are elementary or standard consequences under stated assumptions; their organization around this experimental contract is not a claim of new foundational mathematics. Third, we provide a controlled comparative study with twelve bounded work packages, frozen confirmation, external replication, and an auditable account of what each result supports.

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A. Relation to Prior Work

Predictive stacking combines distributions by optimizing a proper predictive score; Yao *et al.* [1] develop stacking for Bayesian predictive distributions. Its objective and validation access differ from reconstructing a fixed empirical table using released client summaries. Our fitted-model score comparisons are neither integrated Bayesian model evidence nor leave-one-out Bayesian stacking implementations.

Entropy balancing [2] and kernel mean matching [3] establish moment- and embedding-based reweighting. We adapt their calibration viewpoint to scalar weights on client generators. This is distinct from their original observation-weighting settings, but the calibration principle itself is prior work. Finite transport barycenters and entropy-regularized transport also predate this study [4]. A synthetic-support implementation is not an exact continuous barycenter.

Private-PGM reconstructs distributions from statistical measurements using graphical models [5]. AIM adaptively selects and measures statistics for private synthesis [6]. FLAIM extends this line to federation and explicitly addresses heterogeneous clients [7]. HeteroFedSyn studies private horizontal synthesis using distributed marginal selection, including noise-aware dependency estimation [8]. These works prevent a novelty claim based simply on federated synthesis, heterogeneous marginals, or fitting a model to released statistics. Their trust, interaction, and privacy contracts also differ from our nonprivate model-release comparison.

Fed-TGAN includes similarity-aware weighting in a neural federated tabular generator [9]; weighting under heterogeneity is therefore not new in itself. Our non-neural setting studies a more limited question about the value of reweighting *fixed, matched local generators*. Chow–Liu dependency trees [10], copula models, and mixture models are established modelling tools. No new local generator architecture is claimed.

The supported contribution is consequently a reproducible characterization of when calibration fails to deliver a specified practical advantage under a fixed target, together with diagnostics that explain why superficially stronger claims do not follow. Four-cell Private-PGM and adapted FLAIM references provide context, not a state-of-the-art ranking. HeteroFedSyn is discussed but not numerically replicated. A twelve-package completion register documents scope; its length is not evidence of publication novelty.

II. Statistical Setup and Research Objects

A. Records, Clients, and Non-IID Assignment

A record is $Z = (X, Y) \in \mathcal{Z} = \mathcal{X} \times \{0, 1\}$ with a common measurable schema at all K clients. Adult has 14 predictors: six integer-valued numerical attributes and eight categorical attributes, plus the income label. Non-IID denotes unequal client distributions; it does not mean that clients have different columns.

For interpretation only, let P_0 be a source distribution and let measurable assignment probabilities satisfy $a_i(z) \geq 0$ and $\sum_i a_i(z) = 1$. If $I \mid Z = z \sim \mathrm{Categorical}(a(z))$, then

$$\pi_i = \int a_i \, dP_0, \qquad P_i(dz) = \frac{a_i(z)}{\pi_i} P_0(dz), \qquad P_0 = \sum_i \pi_i P_i. \tag{1}$$

for positive π_i. The last identity follows by summing $a_i(z)$. Partitioning on one or two features can induce shifts in feature marginals, labels, and feature–label associations simultaneously. Equation (1) is a population interpretation, not a guarantee that a finite partition behaves as independent population sampling.

The implementation fits the assignment rule on allowed training data, then applies the same rule to all data roles using stable row identifiers for reproducible randomization. The main experiment uses age, age plus education, and label-skew assignments with $K \in \{5, 10\}$. Label-based assignment is a benchmark operation performed by the evaluator, including on held-out labels; it is not a prediction-time routing service.

B. Disjoint Data Roles and a Fixed Empirical Target

Client i has disjoint training, local-tuning, calibration, validation, and assessment data. Denote training size by n_i and fix

$$\alpha_i = \frac{n_i}{N}, \qquad N = \sum_i n_i, \qquad \alpha \in \Delta_K = \{w \in \mathbb{R}_+^K : \mathbf{1}^\top w = 1\}. \tag{2}$$

These are record-count target proportions, not survey expansion weights. Adult's `fnlwgt` remains an ordinary predictor.

For a declared real reference role, write $D_R = \biguplus_i D_i^R$, $t_i = |D_i^R|$, and define its alpha-standardized empirical measure

$$R_\alpha = \sum_i \frac{\alpha_i}{t_i} \sum_{z \in D_i^R} \delta_z. \tag{3}$$

Every client with $\alpha_i > 0$ must have $t_i > 0$. An empty reference client makes this endpoint unsupported; its mass is not redistributed silently. The ordinary pooled empirical measure uses weights $t_i / \sum_j t_j$ and is a different, separately reported reference. Thus D_R is observed, but the primary comparison standardizes its client contributions to a training-defined target.

C. Release Contract and Learned Objects

The locally fitted joint generator is $Q_i \equiv \widetilde{P}_i$. Its release is

$$R_i^{\mathrm{rel}} = (Q_i, M_i), \qquad \Theta = F(R_1^{\mathrm{rel}}, \ldots, R_K^{\mathrm{rel}}; U), \tag{4}$$

where U is server randomness and M_i is declared metadata. Metadata may be public schema information, statistics of the released generator, or additional statistics of private records. The last category is a further data-dependent release. Hiding rows while releasing such information is not differential privacy.

Table I distinguishes the principal objects. The server may sample a model without having an exact probability-mass evaluator. A rounded continuous density and an emitted discrete-row mass are different mathematical objects. Real-reference scoring occurs in an evaluator pathway; its presence does not imply that a deployable server can inspect the same rows.

TABLE I
OBJECTS CONSTRUCTED OR ANALYZED. R_α AND S ARE EMPIRICAL MEASURES; Q_i AND Q_w ARE FITTED-MODEL DISTRIBUTIONS.

Object	Meaning and role
D_i^{tr}, D_i^R	Local fitting records and held-out reference records.
α, w	Fixed target proportions and chosen generator weights.
Q_i, M_i	Released joint model and permitted metadata.
$Q_w = \sum_i w_i Q_i$	Aggregate mixture; nonlinear pools are separate operators.
D_S, S	Generated table and its normalized empirical measure.
$\mathcal{Q}, b_R, b_S$	Finite workload and real/synthetic answer vectors.
$H, \widehat{H}, \widehat{b}$	Model answers, pilot estimates, and calibration answers.
$\mathcal{W}, \widehat{w}, w^*$	Feasible representation set, chosen weights, finite-object oracle.
$Q_w^X(\cdot \mid C)$	Aggregate-implied conditional feature law.
M_C	Separately fitted per-condition student model.
η_{Q_w}, g_S	Joint-implied probability and synthetic-trained predictor.
$B, T, \varepsilon, \delta$	Communication, runtime, and conditional privacy accounting.

D. Empirical Fidelity and Representation

For fixed $\mathcal{Q} = \{q_1, \ldots, q_L\}$, $q_\ell : \mathcal{Z} \to [0,1]$, let

$$b_{R,\ell} = \sum_i \frac{\alpha_i}{t_i} \sum_{z \in D_i^R} q_\ell(z), \tag{5}$$

$$b_{S,\ell} = \frac{1}{m} \sum_{z \in D_S} q_\ell(z), \qquad S = \frac{1}{m} \sum_{z \in D_S} \delta_z, \tag{6}$$

$$\mathcal{L}_{\mathcal{Q}}(D_S, D_R; \alpha) = \|b_S - b_R\|_\infty . \tag{7}$$

The primary workload has $L = 152$ queries: 143 one-way queries, eight additional two-way queries, and one structural education-code query. Twenty-three additional queries form a viewed-development secondary audit. They are not an independent post-development discovery set.

The paired contrast is

$$\Delta = \mathcal{L}_{\mathcal{Q}}(D_S^{\mathrm{challenger}}, D_R; \alpha) - \mathcal{L}_{\mathcal{Q}}(D_S^{\mathrm{size}}, D_R; \alpha). \tag{8}$$

Negative values favour the challenger. The registered practical margin is $\Delta \le -0.005$ after averaging draws within cells and seeds within a declared slice. Changing w never changes α or the comparison reference.

The retained constrained challenger uses

$$\mathcal{W} = \{w \in \Delta_K : \|w - \alpha\|_1 \le 0.25,\ w_i \ge 0.5\alpha_i \quad \forall i\} . \tag{9}$$

The set contains α. We additionally record subgroup prevalence, coverage, and minimum w_i/α_i. A client-mass constraint does not certify fairness or accurate within-client dependencies. An effective contribution count, $1/\sum_i w_i^2$, is descriptive and does not count independent institutions or people.

E. Finite-Workload Versus Population Statements

For any measures A, B, define $d_{\mathcal{Q}}(A,B) = \max_\ell |\mathbb{E}_A q_\ell - \mathbb{E}_B q_\ell|$. This is a pseudometric: different distributions may agree on every listed query. The measured endpoint is exactly $d_{\mathcal{Q}}(S, R_\alpha)$. Total variation, $\mathrm{TV}(A,B) = \sup_E |A(E) - B(E)|$, upper bounds $d_{\mathcal{Q}}$, but no converse follows for this workload. We use arbitrary measures in proofs to state *conditional mathematical implications*; we do not report unobserved population TV as a measured performance quantity.

III. GENERATORS, AGGREGATION, AND EVALUATION PROCEDURES

A. Non-Neural Local Generators

The corrected core compares three families. A Gaussian copula fits empirical marginal transformations and Gaussian latent dependence. Numerical ties and thresholds for categorical levels make its score approximate. Nominal categories receive a deterministic order; dependence estimates can change under a different ordering. A Chow–Liu generator fits a dependency tree to discretized variables with smoothed conditional tables, then samples integer values within numerical bins. Its declared integer/categorical probability mass includes the within-bin correction. A mixed Gaussian mixture combines transformed numerical Gaussian components with component-conditional categorical probabilities. Its inverse-transformed, rounded, clipped outputs do not have the same mass function as its continuous scoring density.

Three independent-histogram cells form a weak-model control. Prototype-kernel and latent-class implementations belong to earlier exploratory work and are not part of the 54-cell corrected core. The paper does not pool their pilot outcomes with the main comparison.

For the frozen copula, each client fits six settings: bins in $\{8, 16, 32\}$ crossed with smoothing parameter $a \in \{0.1, 1\}$. Here a is a model parameter and is unrelated to client target α. Each fit uses only its client's training rows. Local tuning selects the lowest mean absolute primary-query discrepancy between 1,000 generated rows and local-tune records, with the earlier grid entry breaking an exact tie. Eligibility requires the recorded support and convergence checks, valid income labels, and finite scores on the first 64 local-tune records. No global validation or test row extends the grid.

B. Deterministic Schema Projection

Let T set education-number to the public code corresponding to a recognized education category. Unmapped categories remain explicit rather than receiving invented values. All corrected generators and their pilots emit the pushforward $T_\# Q_i$. We henceforth absorb this transformation into Q_i. The structural query counts both unmapped education and code disagreement as violations; therefore zero known-category mismatch is compatible with a small nonzero overall structural violation rate.

The same transformation applies to size and alternative rules. It repairs a known relation, not every dependence in Adult. Exact finite-model scoring requires summing over all preimages under T; the discrete audit uses the 17 possible raw education-number values in the declared 0–16 support. Applying that sum to an approximate copula or GMM score does not turn it into an exact mass function.

C. Weighting Families and Their Information Requirements

Let $H_{\ell i} = \mathbb{E}_{Q_i} q_\ell$ and let $\widehat{H}$ be estimated from synthetic pilots. Let $\widehat{b}$ be an alpha-weighted vector of released, unnoised real calibration query means. In the main constrained fit, the pilot has 1,500 rows per client. The following objectives clarify the implemented families and their boundaries.

a) Size and uniform baselines: Size weighting uses $w = \alpha$; uniform weighting uses $w_i = 1/K$. Uniform weighting targets equal contributions only when that is the declared target. It is not an automatically fair correction for a naturally unequal record population.

b) Predictive-score and evidence rules (F1–F2): On a common real calibration reference z_j, let ℓ_{ji} be the supplied log score and $\omega_j > 0$ be reference weights scaled to sum to the number of records. Write $s_i = \sum_j \omega_j \ell_{ji}$. A predictive-score softmax has $w_i \propto \exp(s_i/\tau)$; a conditional predictive-evidence rule has $w_i \propto \pi_i^{\text{prior}} \exp(s_i)$. These are scores of already fitted candidates. They do not integrate over training-model parameters; weighted observations give a power-score calculation, not ordinary Bayesian evidence. Their real-row score access is privileged.

c) Stacking (F3): The fitted-model objective is

$$\max_{w \in \Delta_K} \sum_j \omega_j \log \left(\sum_i w_i \exp(\ell_{ji}) \right). \tag{10}$$

It measures predictive fit on that reference, not primary query-max fidelity. Exact and approximate scores are labelled, and raw scores are not ranked across incompatible model families.

d) Moment calibration (F4): Least-squares calibration minimizes $\left\| \widehat{H} w - \widehat{b} \right\|_2^2$. Entropy regularization or residual covariance whitening yields supporting variants, respectively involving $\mathrm{KL}(w \| \alpha)$ or $(\widehat{H} w - \widehat{b})^\top \Sigma^{-1} (\widehat{H} w - \widehat{b})$. Whitening concerns uncertainty in a common residual; it is not inverse-variance averaging of clients assumed to share an identical truth.

e) Kernel matching (F5): The same simplex-constrained quadratic solver matches released kernel feature means. Its inputs must actually be embeddings; a generic moment matrix is not, by renaming, a characteristic-kernel guarantee. A finite feature embedding does not identify arbitrary distributions.

f) Robust calibration (F6): A query-discrepancy objective with a recorded uncertainty-radius term explores robustness to summary error. This is an implementation-specific robust calibration variant, not a faithful reproduction of every distributionally robust federated-learning method. The retained challenger is the simpler constrained minimax program

$$\begin{aligned} \min_{w,t,u} \quad & t \\ \text{s.t.} \quad & -t\mathbf{1} \le \widehat{H} w - \widehat{b} \le t\mathbf{1}, \quad w \in \Delta_K, \\ & -u_i \le w_i - \alpha_i \le u_i, \quad u_i \ge 0, \\ & \sum_i u_i \le 0.25, \quad w_i \ge 0.5 \alpha_i. \end{aligned} \tag{11}$$

HiGHS solver status and feasibility certificates apply to this finite matrix. A validation-vector version is a privileged fixed-pilot oracle diagnostic. Neither is a population oracle.

TABLE II
ACCESS CLASSES. NO MAIN BTP ROW HAS AN END-TO-END DP GUARANTEE.

Class	Server-side inputs	Interpretation
Size/uniform	Models, schema, declared counts	Direct release-only baselines.
Summary rules	Models and unnoised calibration summaries	Additional nonprivate information.
F1–F3	Common real-row scores	Privileged reference access.
Validation LP	Model pilot and real validation answer vector	Finite privileged benchmark.
Private-PGM reference	Same released unnoised summaries	Nonprivate measurement fitting.
Adapted FLAIM	Interactive raw-client marginal access	Different trust/privacy contract.
TRTR predictor	Real train/tune/calibration records	Privileged utility reference.

g) Geometric pooling (F7): On a finite synthetic candidate support $\mathcal{S}$, a geometric pool assigns mass proportional to $\exp\{\sum_i w_i \log s_i(z)\}$. The score convention and floor for impossible scores must be declared. This is a different aggregation operator from $\sum_i w_i Q_i$, and its support and smoothing are part of the method.

h) Entropic barycenters (F8): Finite synthetic supports and a declared mixed-data transport cost define an entropy-regularized transport pool. Its recorded finite weight search does not solve the general continuous barycenter problem. Approximate posterior smoothing over its candidate rows is a separate prediction route, not the conditional probability of its finite joint mass.

These eight families, together with supporting baselines, were research candidates. They are not eight new inventions. The corrected source matrix contains 13 method rows per cell; the separately audited constrained rule is the prespecified challenger carried into confirmation. Table II preserves the differences in access.

D. Generation, Conditional Sampling, and Prediction

For a desired table of m rows, largest-remainder allocation produces integer counts m_i summing exactly to m, with $|m_i/m - w_i| < 1/m$. Independently generated client rows are combined and shuffled. This is a stratified allocation, not IID categorical sampling of a client for every row. Five draws at $m = 10{,}000$ are used in the main comparisons.

For a condition $C \subseteq \mathcal{X}$, the frozen service first draws a 30,000-row synthetic teacher support pilot. It proceeds only if at least 300 rows and at least 1% of that pilot satisfy C. Direct rejection then requests 1,000 accepted feature rows with at most 200,000 proposals. Conditions are age 25–45, female, bachelor's education with age 30–60, and doctorate education with age 60–90. Public-schema validation rejects impossible requests. The support gate is operational, not a confidence certificate for real conditional mass. Separate per-condition Chow–Liu students were evaluated during development and were not selected for the frozen service.

Income predictors use logistic regression and extra-trees with source-specific tuning and Platt calibration. Logistic $C \in \{0.1, 1, 10\}$ and extra-trees minimum leaf size in $\{3, 8\}$ are selected by tune Brier score; extra-trees uses 120 estimators. The same source supplies a separate calibration set. Probabilities are clipped to $[10^{-6}, 1-10^{-6}]$ for log loss. TSTR uses 10,000/2,000/2,000 generated train/tune/calibration rows. Privileged TRTR uses 21,148/4,880/3,254 real rows in confirmation. These unequal budgets are part of the comparison.

IV. Guarantees and Boundaries

This section proves the results needed to interpret the experiments. All inequalities distinguish finite observations, fitted-model quantities, and sampling assumptions. The analytical fallback below was developed from the completed study and was *not* the campaign's selection algorithm. Its existence does not retroactively certify any reported comparison.

A. Mixture Geometry and Representation

For the fixed real answer vector b_R, define the model-answer objective

$$L_R(w) = \|Hw - b_R\|_\infty . \tag{12}$$

It is a computational intermediary for analyzing the generated-table loss; H may not be exactly available. Since $w \mapsto Hw$ is linear,

$$\inf_{w \in \Delta_K} L_R(w) = \operatorname{dist}_\infty\big(b_R, \operatorname{conv}\{H_{\cdot i}\}_{i=1}^K\big). \tag{13}$$

Over $\mathcal{W}$, the achievable set is the smaller polytope $H\mathcal{W}$. A query target outside it cannot be recovered by scalar reweighting of the fixed generators.

Proposition 1 (Sharp representation bound)**.** *For $H \in [0,1]^{L\times K}$ and $w, \alpha \in \Delta_K$, let $D_H = \max_{i,j} \|H_{\cdot i} - H_{\cdot j}\|_\infty$. Then*

$$\|H(w-\alpha)\|_\infty \le \frac{D_H}{2}\|w-\alpha\|_1 \le \frac{1}{2}\|w-\alpha\|_1 . \tag{14}$$

The constant $1/2$ is attained by some two-client query matrices.

Proof. Let $v = w - \alpha$ and $s = \sum_{v_i>0} v_i = \frac{1}{2}\|v\|_1$. For $s > 0$, its positive and negative parts, divided by s, are probability vectors. A row h therefore satisfies $|hv| \le s(\max_i h_i - \min_i h_i)$. Maximize over rows. For sharpness, take $K = 2$, $h = (0,1)$, $\alpha = (1,0)$, and $w = (1-t, t)$. Both sides of the last inequality equal t. For a fixed row, equality requires its extrema on the respective positive and negative supports of v. □

At budget 0.25 this bounds query change relative to the same size-weighted models by 0.125, not by the much smaller practical margin 0.005. It does not bound error relative to real data. It is attained on distinguishable clients and can be zero for identical client answers despite a large weight change.

Proposition 2 (Observable error decomposition)**.** *Let $B_{\ell i}$ be the empirical query mean in D_i^R, so $b_R = B\alpha$. Then*

$$\|b_S - B\alpha\|_\infty \le \|b_S - Hw\|_\infty + \sum_i \alpha_i \|H_{\cdot i} - B_{\cdot i}\|_\infty + \frac{D_H}{2}\|w-\alpha\|_1 . \tag{15}$$

For an additional aggregation operator or compressed model, add the discrepancy between its query-answer vector and Hw.

Proof. The exact vector identity is $b_S - B\alpha = (b_S - Hw) + (H-B)\alpha + H(w-\alpha)$. Apply the triangle inequality, convexity, and Proposition 1. An operator approximation is another inserted vector. The components need not be independent, positive contributions: errors may cancel before norms are taken. □

Thus the decomposition is an upper bound, not a causal allocation of observed error. Replacing H by a finite pilot gives an exact empirical telescoping identity with pilot noise inside its terms.

Proposition 3 (Subset collapse and identifiability)**.** *If a subset generator is $Q_B = \sum_{i\in B} w_i^{(B)} Q_i$ and subsets are mixed with weights ρ_B, then*

$$\sum_B \rho_B Q_B = \sum_i \omega_i Q_i, \qquad \omega_i = \sum_{B \ni i} \rho_B w_i^{(B)}. \tag{16}$$

Regrouping introduces no new mixture distributions. Query answers identify all simplex weights uniquely if $\ker H \cap \{v : \mathbf{1}^\top v = 0\} = \{0\}$; otherwise feasible null directions can make different weights observationally indistinguishable.

Proof. Exchange the finite sums; nonnegativity and normalization give $\omega \in \Delta_K$. Equality of two answer vectors implies $H(w-w') = 0$ and $\mathbf{1}^\top(w-w') = 0$. The kernel condition is sufficient and necessary for injectivity on the whole simplex because any nonzero zero-sum direction can be scaled around an interior point. Boundary constraints can yield uniqueness at a particular point without global injectivity. □

Preserving latent client contribution exactly requires $\omega = \alpha$. Distributional equality alone need not require it when components are redundant. A regularizer can choose unique weights without making the underlying target identifiable.

Counterexample 4 (Model-only information is insufficient)**.** Suppose identical releases in two possible worlds are $Q_1 = \text{Bernoulli}(1/4)$ and $Q_2 = \text{Bernoulli}(3/4)$. No assumption connects model error to the real reference. For $q(z) = z$, a real empirical target with fraction $1/4$ of ones has oracle $w = (1,0)$, while fraction $3/4$ has oracle $w = (0,1)$. A release-only rule returns the same answer x in both worlds. At least one error is $\max\{|x - 1/4|, |x - 3/4|\} \ge 1/4$. Randomization cannot remove this worst-world expected lower bound, since the two absolute errors sum to at least $1/2$ pointwise. Informative summaries or modelling assumptions are necessary.

B. Finite-Pilot Oracle and Stability

Let $\widehat{b}$ be the permitted calibration vector and assume the deterministic error event

$$\max_i \left\|\widehat{H}_{\cdot i} - H_{\cdot i}\right\|_\infty \le r_H, \qquad \left\|\widehat{b} - b_R\right\|_\infty \le r_b. \tag{17}$$

Set $r = r_H + r_b$ and $\widehat{L}(w) = \left\|\widehat{H}w - \widehat{b}\right\|_\infty$.

Theorem 5 (Uniform perturbation and oracle inequality)**.** *Uniformly over every $w \in \Delta_K$,*

$$|\widehat{L}(w) - L_R(w)| \le r. \tag{18}$$

Let $\Omega(w) = \frac{1}{2}\|w - \alpha\|_2^2$ and let $\widehat{w}_\lambda \in \mathcal{W}$ minimize $\widehat{L}(w) + \lambda\Omega(w)$ to tolerance $\zeta \ge 0$. If $w^* \in \arg\min_{w\in\mathcal{W}} L_R(w)$, then

$$L_R(\widehat{w}_\lambda) - L_R(w^*) \le 2r + \lambda\Omega(w^*) + \zeta \le 2r + \lambda + \zeta. \tag{19}$$

This is valid for data-adaptive weights chosen from the same finite pilot.

Proof. For simplex w, $\left\|(\widehat{H} - H)w\right\|_\infty \le r_H$. The reverse triangle inequality gives (18). Compare the approximate regularized minimizer with w^*, use (18) twice, and drop $-\lambda\Omega(\widehat{w}_\lambda) \le 0$. Since two simplex points have squared distance at most two, $\Omega \le 1$. The argument is uniform in w and therefore covers selection of weights from $\widehat{H}, \widehat{b}$; it does not cover arbitrary data-adaptive query discovery without an additional uniform event. □

If r_b is unknown, the bound is not a certificate. A smaller optimization tolerance only controls ζ and cannot eliminate pilot or target uncertainty. The unregularized experiment has $\lambda = 0$; the regularized family is analyzed to explain stability, not reported as the frozen winner.

Proposition 6 (Regularization stability)**.** *For two objectives F, F' that share the term $\lambda\Omega$, $\lambda > 0$, suppose their remaining terms are convex on the same convex feasible set and $\sup_w |F(w) - F'(w)| \le \eta$. Their unique exact minimizers w, w' obey*

$$\|w - w'\|_2 \le 2\sqrt{\eta/\lambda}. \tag{20}$$

Proof. Strong convexity and constrained optimality give $\frac{\lambda}{2}\|w - w'\|_2^2 \le F(w') - F(w)$. Adding and subtracting F' and using optimality of w' bounds the right side by 2η. Strong convexity also gives uniqueness. Stability increases with λ, at the cost of the comparator term in (19). □

C. Synthetic Allocation and a Sufficient Fallback Certificate

Theorem 7 (Finite-workload stratified sampling)**.** *Condition on fitted models and selected weights. Generate m_i independent rows from Q_i, with $\sum_i m_i = m$, and write $\bar{w}_i = m_i/m$. For fixed L bounded queries, with probability at least $1 - \gamma$,*

$$\|b_S - Hw\|_\infty \le u_m(\gamma) + \rho_m(w), \quad u_m(\gamma) = \sqrt{\frac{\log(2L/\gamma)}{2m}}, \tag{21}$$

where $\rho_m(w) = \frac{1}{2}\|\bar{w} - w\|_1$. Largest-remainder allocation gives $\rho_m(w) < K/(2m)$. The same u_m applies to IID mixture sampling with $\rho_m = 0$.

Proof. Each query average is a mean of m independent variables in $[0, 1]$, not necessarily identically distributed. Its expectation is $H\bar{w}$. Hoeffding's exponential argument yields tail probability at most $2e^{-2mu^2}$ per query. A union bound gives the displayed u_m. Proposition 1 bounds $\|H(\bar{w} - w)\|_\infty$ by ρ_m. Each rounded count differs from mw_i by less than one. Shuffling does not change query averages. □

In particular, $|\mathcal{L}_\mathcal{Q}(D_S, D_R; \alpha) - L_R(w)| \le u_m + \rho_m$. More generated rows reduce this sampling term, not the underlying fitted-model objective. A desired sampling term u for IID draws is ensured by $m \ge \log(2L/\gamma)/(2u^2)$.

Theorem 8 (Sufficient fallback certificate)**.** *On event* (17)*, choose any candidate $\widehat{w} \in \mathcal{W}$ using the released pilot and summaries. Let $\kappa > 0$ be a desired gain. Accept only when*

$$\widehat{L}(\alpha) - \widehat{L}(\widehat{w}) \ge \kappa + 2r + 2u_m(\gamma_S/2) + \rho_m(\alpha) + \rho_m(\widehat{w}). \tag{22}$$

Generate fresh candidate and size tables independently of selection. With conditional probability at least $1 - \gamma_S$, acceptance guarantees

$$\mathcal{L}_\mathcal{Q}(D_S^{\widehat{w}}, D_R; \alpha) \le \mathcal{L}_\mathcal{Q}(D_S^{\alpha}, D_R; \alpha) - \kappa. \tag{23}$$

On rejection, returning the already generated size table has exactly its loss. If event (17) *holds with probability $1 - \gamma_0$, the joint guarantee is at least $1 - \gamma_0 - \gamma_S$.*

Proof. Uniform perturbation gives $L_R(\widehat{w}) - L_R(\alpha) \le \widehat{L}(\widehat{w}) - \widehat{L}(\alpha) + 2r$. Apply Theorem 7 to the two fresh tables and use a union bound. The sum of their deviations is the last three terms in (22). Returning the same baseline table on rejection avoids falsely promising equality between two independent baseline draws. □

This is a conservative sufficient condition. It may never accept. With $L = 152$, $K = 10$, 1,500 pilot rows per client, and pilot failure budget 0.01, the elementary union bound gives $r_H \simeq 0.0649$, before real-summary or final-draw uncertainty. Twice this radius already exceeds 0.005 substantially. This is a worst-case analytical illustration, not a retrospective estimate of the campaign's actual error or a claim that improvement is impossible. Appendix A states assumptions under which r_b can be certified.

D. Conditionals and Prediction Need Additional Control

For $C \subseteq \mathcal{X}$, define $c_i(C) = Q_i^X(C)$ and $c_w(C) = \sum_i w_i c_i(C)$. When $c_w(C) > 0$,

$$Q_w^X(\cdot \mid C) = \sum_i \frac{w_i c_i(C)}{c_w(C)} Q_i^X(\cdot \mid C). \tag{24}$$

Zero-mass components contribute zero. This follows by dividing $\sum_i w_i Q_i^X(B \cap C)$ by $c_w(C)$. Routing conditioned requests with the original w_i is generally incorrect. The corresponding *real empirical* conditional reference is $R_\alpha(\cdot \mid C)$; it likewise uses condition-dependent client responsibilities, not an unadjusted average of client conditionals.

Proposition 9 (Empirical conditional transfer)**.** *For two fixed measures R, S with $r = R(C) > 0$, $s = S(C) > 0$, let $e_C = \max_{q\in\mathcal{Q}_C} |\mathbb{E}_S[\mathbf{1}_C q] - \mathbb{E}_R[\mathbf{1}_C q]|$ and $e_0 = |s - r|$. Then*

$$d_{\mathcal{Q}_C}(S(\cdot \mid C), R(\cdot \mid C)) \le \min\{1, (e_C + e_0)/r\}. \tag{25}$$

Proof. For $a = \mathbb{E}_R[\mathbf{1}_C q]$ and $c = \mathbb{E}_S[\mathbf{1}_C q]$, add and subtract c/r in $|c/s - a/r|$. The result is at most $|c - a|/r + c|r - s|/(rs) \le (|c - a| + |r - s|)/r$, since $c \le s$. Take the maximum and cap by one. If $s = 0$, the synthetic conditional is undefined and must abstain. □

The workload must contain or separately measure $\mathbf{1}_C$ and $\mathbf{1}_C q$. Small unconditional errors on unrelated queries do not supply e_C. A teacher/student comparison satisfies the triangle inequality only on the same conditional workload and reference. The two finite samples' discrepancy includes their sampling variation, not solely student approximation error.

Counterexample 10 (Matching marginals can reverse a conditional)**.** Let R put mass $1/2$ at $(X,Y) = (0,0), (1,1)$ and S at $(0,1), (1,0)$. Their one-way means agree exactly. Nevertheless $R(Y = 1 \mid X = 1) = 1$ and $S(Y = 1 \mid X = 1) = 0$. A perfect synthetic predictor $g(x) = 1 - x$ has real squared loss one. Thus neither conditional fidelity nor predictive utility follows from these marginal matches. Higher-order parity distributions give analogous failures for workloads omitting the relevant interaction.

For a finite exact joint model with positive denominator, its implied predictor is

$$\eta_{Q_w}(x) = \frac{\sum_i w_i q_i(x,1)}{\sum_i w_i \{q_i(x,0) + q_i(x,1)\}}. \tag{26}$$

Equivalently it averages local conditionals with feature-dependent responsibilities $w_i q_i^X(x) / \sum_j w_j q_j^X(x)$. This Bayes identity is not a learned feature-adaptive gate and does not establish the originally proposed shared/private novelty. A separately trained g_S is a distinct object. Appendix C proves the relevant bounded-loss transfer statements and explains why the 152-query metric alone cannot certify them.

V. Experimental Design

A. Data Integrity and Development History

UCI Adult provides 32,561 official training records and 16,281 official test records [11]. Whitespace and terminal periods in test labels are normalized; missing `?` values remain explicit. Stable row IDs are metadata, never predictors. Twenty-five official-training rows exactly matching complete official-test records were quarantined. The remaining exact duplicate groups stay together across income-stratified development roles. The original development allocation is approximately 65% train, 15% local tune, 10% calibration, and 10% validation; the validation pool is further separated into selection-validation and development-assessment roles in development comparisons. Client assignment is training-fitted and reused unchanged.

The official test was read earlier for source validation and duplicate quarantine. It was first *scored for confirmation* after the complete written freeze. We do not describe it as never physically read before that time. Development diagnostics informed structural projection, workload version 2, zero-atom handling, and support policies. Historical unprojected results and the earlier audit-based tie break are therefore exploratory. They remain archived rather than being silently overwritten or merged with the corrected matrix.

All experiments simulate clients, server, and evaluator on one EC2 host with one BTP worker and one native numerical thread. Access boundaries are recorded by interfaces and release ledgers; they are not cryptographic isolation or a deployed multi-institution network.

B. Matrices, Selection, and Analysis Units

The corrected core comprises 3 generators $\times$ 3 partition types $\times$ 2 client counts $\times$ 3 development seeds (101, 202, 303), giving 54 fitting cells. Three independent-histogram controls are separate. Across the 57 source cells there are 741 method rows, 3,705 method/cell/draw assessment rows, and 57 successful fixed-pilot LP records. Repeated analyses reuse checkpoints when source, split, workload, and sampling hashes agree. Auxiliary constrained evaluations are recorded separately from the 13-row source matrix.

A cell first averages its five synthetic-draw losses. An Adult development slice averages three cell means for one generator/partition/K combination; confirmation averages two seeds (404, 505) for one partition/K combination. An overall result averages the declared cells equally. A pooled count of all generated rows is not substituted for these aggregation units. The split seeds reuse people, as do all confirmation configurations on the official test. We report descriptive paired means and seed ranges, not independent-cohort p-values or equivalence intervals.

Final family selection used five compatible validation draws for each of 18 cells per family. Size-weighted primary means were 0.029975 for copula, 0.031455 for Chow–Liu, and 0.051446 for GMM. The absolute 0.001 tie tolerance did not apply; specified byte, runtime, and lexical tie breaks were not invoked. This selected copula globally. It did not select a different generator in each test cell. Earlier design was assessment-informed, although this final family comparison used validation only.

The immutable global freeze fixed generator, local tuning, preprocessing, target, size weights, constrained/uniform comparators, workload, exact output size, seeds, conditional support, and predictive calibration. Its UTC timestamp is 2026-10-25 05:39:46.520200; first logged confirmation scoring was 46.553 seconds later. Frozen confirmation contains 12 fitting cells and 180 synthetic tables: three rules and five draws per cell. No confirmation result is used to replace the selected procedure.

C. External Replication

The Bank Marketing replication uses `bank-additional-full.csv`, 41,188 records, rather than the older 45,211-row variant [12]. Post-contact duration is removed before modelling and workload construction. Unknown categories are retained, `pdays=999` is represented as a categorical no-prior-contact atom, and economic decimal attributes are reversibly scaled by 1,000 to integers. Support bounds and vocabularies use training data only and remain unnoised metadata.

A duplicate-group-aware random split at seed 7171 creates approximately 55/15/10/10/10% train/tune/calibration/validation/assessment roles; the assessment has 4,119 rows. Groups are defined after dropping duration. The source is chronological, but this evaluation is not a chronological split. Person IDs are unavailable and contacts can repeat; group handling does not establish person independence.

TABLE III
REGISTERED FIXED-TARGET COMPARISON. NEGATIVE DIFFERENCES FAVOUR CONSTRAINED WEIGHTING. PRACTICAL-GAIN COUNTS USE THE DECLARED SLICE MEANS.

Evidence	Cells	Mean Δ	Gain slices
Adult development	54	+0.002428	0/18
Adult confirmation	12	-0.000215	0/6
Bank replication	12	-0.001332	0/4

Margin: $\Delta \leq -0.005$. Adult development uses three seeds per slice; Adult confirmation uses two; Bank uses three. These counts are not independent significance tests.

Copula and the Adult-validation runner-up Chow–Liu are transferred to age-plus-education and label-skew assignments, $K = 5$, seeds 611/622/633: 12 fitting cells. The registered six-setting local grid, fixed representation constraints, 10,000-row draws, and five repeats are retained. The Bank synthetic pilot uses 5,000 rows per client. Its training-defined workload and all assessment choices were frozen prospectively. The 0.005 margin is transferred as an operational criterion, not assumed to have identical substantive meaning for every dataset.

VI. RESULTS

A. Primary Fixed-Target Comparison: Experiments 1 and 2

Table III summarizes the primary comparison. In corrected Adult development, constrained weighting has mean $\Delta = +0.002427903$ and zero practical-gain slices among 18. Generator-level mean differences are +0.001296 for Chow–Liu, +0.001976 for copula, and +0.004012 for GMM. Three individual constrained cells meet the margin, but none of their registered three-seed slices does. This distinction prevents individual favourable cells from becoming the main conclusion.

A within-cell validation-selected release-only rule, unconstrained in the final representation sense, yields mean $\Delta = -0.000458$ over the core. Only GMM/label-skew/$K = 10$ meets the slice margin, at -0.006951, but its mean ℓ_1 mass shift is 1.222, well above 0.25. On the same models the constrained result is -0.001746. The selected slice therefore does not establish an acceptable fixed-target gain under the declared budget. It remains a recorded trade-off, not a suppressed positive result.

Table IV reports all 13 source candidates across the 54 corrected cells. It does not make them equal-access or equal-capacity comparisons. In particular, finite nonlinear operators, privileged score rules, and unconstrained calibration are diagnostic families. Their results cannot substitute for the retained constrained comparison.

All 57 fixed-pilot LPs succeeded. A certificate proves that the optimizer solved its finite query matrix, objective, target vector, and constraints. It says nothing about the unobserved model expectation matrix, new generated rows, all queries, or nonlinear pooling optima. In six repeated-pilot cells, pilots of 500, 1,500, 5,000, and 15,000 rows/client were each repeated five times; an independent 30,000-row/client matrix supplied a finite reference. At 15,000, mean same-target reference-matrix LP excess was 0.000832 for Chow–Liu age/$K = 5$ and 0.000878 for GMM label-skew/$K = 10$. Their fresh-table differences were +0.000262 and +0.005894. Improved pilot approximation did not establish practical fresh-sample gain.

B. Frozen Adult and Bank Comparisons

The frozen Adult size procedure has mean primary loss 0.022215720. Constrained weighting changes it by -0.000214907, with zero of six partition/K slices reaching -0.005. Uniform weighting changes it by +0.041139 overall and is particularly adverse in label-skew cells. Table V and Fig. 2 retain every slice rather than reporting only the overall mean.

The external Bank mean difference is -0.001331749 with zero practical-gain slices among four. Chow–Liu label skew has the largest favourable slice, -0.004008, still short of the registered margin; copula age-plus-education is adverse by +0.000441. These results support the bounded negative comparison on a second dataset. They neither prove equivalence nor exclude gains from other local model families, workloads, access budgets, or representation policies.

C. Subset and Sampling Results: Experiments 3 and 4

Experiment 3 closes as an algebraic negative: regrouping mixtures of overlapping client subsets reduces to effective weights by Proposition 3. A numerical identity check is consistent with the proof but is not itself its proof. No new within-subset estimator was implemented, so an expanded expressive family is not claimed.

Experiment 4 reuses six fixed checkpoints and weights at $m \in \{2{,}500, 10{,}000, 40{,}000\}$, with five draws each. For size weighting, the mean five-draw loss range falls from 0.009476 to 0.002542 in Chow–Liu age/$K = 5$, and from 0.015467 to 0.003472 in GMM label-skew/$K = 10$. At 40,000 rows their constrained-minus-size differences are +0.006713 and +0.001492. None of the six family/size means meets the practical margin. The remaining error is an empirical floor over these checkpoints and tested sizes, not an estimated population lower bound. Some sizes may share random-stream prefixes.

D. Conditional Synthesis: Experiment 5

Development compared independently trained per-condition Chow–Liu students with direct teacher rejection on six projected cells, matching the number of accepted rows between routes. For age and bachelor's-plus-age conditions, direct rejection had lower family/weight mean query-max error than the student. Raw student filter violations for these requests were approximately 20–24%; postfilter samples had zero recorded condition and mapped education-code violations. Postfilter validity must not be confused with the raw sampler's acceptance rate.

The frozen direct service completed all three common conditions in every Adult confirmation cell, with 1,000 accepted rows and zero filter violations. Mean conditional feature-query maxima were 0.083787 for age 25–45, 0.147013 for bachelor's plus age 30–60, and 0.375000 for female. The rare older-doctorate request abstained in all 12 cells. These abstentions remain explicit outcomes.

The female result is a severe dependency failure. In all 12 cells the worst query was `relationship=Husband`; its

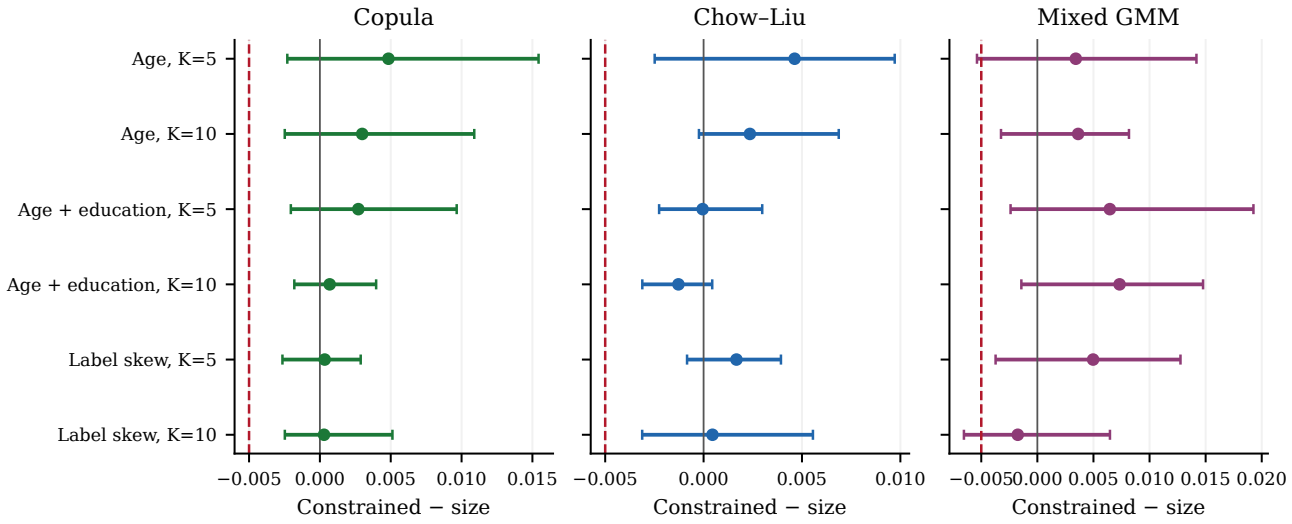


Fig. 1. Corrected Adult development: constrained-minus-size primary query-max difference for each three-seed slice. Points are means; horizontal ranges are the minimum and maximum seed-level paired means, *not confidence intervals*. Five draws were averaged within each seed. The dashed line is the registered practical margin. All 18 slice means remain above it.

TABLE IV
ALL SOURCE CANDIDATE METHODS OVER THE 54 CORRECTED ADULT CORE CELLS; DESCRIPTIVE ASSESSMENT MEANS.

Method	Access class	Query-max	Δ vs. size	Mean $\|w-\alpha\|_1$
Size	Models/counts	0.035633	+0.000000	0.000
Uniform	Models/counts	0.049719	+0.014086	0.391
Moment least squares	Models/summaries	0.034868	-0.000764	0.590
Entropy calibration	Models/summaries	0.035162	-0.000471	0.152
Residual GLS	Models/summaries	0.035031	-0.000601	0.559
Kernel mean matching	Models/summaries	0.035868	+0.000235	0.738
Unconstrained query minimax	Models/summaries	0.039671	+0.004038	0.969
Robust query calibration	Models/summaries	0.037909	+0.002277	0.806
Predictive-score softmax	Privileged scores	0.134920	+0.099288	1.588
Predictive evidence	Privileged scores	0.134920	+0.099288	1.588
Stacking	Privileged scores	0.049619	+0.013986	0.837
Finite geometric pool	Models/summaries	0.298326	+0.262693	1.441
Finite entropic barycenter	Models/summaries	0.103013	+0.067381	0.414

Each method/cell averages five draws. The three independent-histogram controls are excluded. The representation-constrained challenger is a separately audited variant (Table III). For nonlinear pools the last column refers to their recorded input weights, not an equivalent output client-mixture mass. Privileged scores, finite nonlinear supports, and different released summaries prevent an equal-information ranking. These are implementation outcomes, not optimal values for each literature family.

TABLE V
FROZEN ADULT CONFIRMATION: FIVE DRAWS PER CELL, TWO FITTING SEEDS PER SLICE.

Partition	K	Size loss	Δ_C	Δ_U
Age	5	0.025883	+0.000043	-0.001019
Age	10	0.026583	-0.000156	+0.042807
Age + education	5	0.022456	-0.000579	+0.000420
Age + education	10	0.022922	-0.000823	-0.000241
Label skew	5	0.018286	+0.000276	+0.142920
Label skew	10	0.017164	-0.000050	+0.061946

Δ_C: constrained minus size; Δ_U: uniform minus size. All configurations use the same 16,281 official reference records.

TABLE VI
BANK EXTERNAL REPLICATION: $K = 5$, THREE FITTING SEEDS AND FIVE DRAWS PER CELL.

Generator	Partition	Size loss	Δ_C
Chow–Liu	Age + education	0.044545	-0.000547
Chow–Liu	Label skew	0.046878	-0.004008
Copula	Age + education	0.049572	+0.000441
Copula	Label skew	0.051488	-0.001213

alpha-standardized real female-conditional frequency was zero in the reference records, whereas generated frequency ranged from 0.340 to 0.399. Figure 4 exposes this result directly. It is an observed defect of this model on this reference, not a universal demographic rule. Passing the condition filter or the synthetic support gate did not guarantee the distribution of remaining attributes.

E. Utility and Joint Prediction: Experiments 6 and 7

Table VII reports same-person, alpha-weighted confirmation metrics. Size-weighted synthetic logistic training yields Brier 0.129818 and AUROC 0.856167, compared with privileged

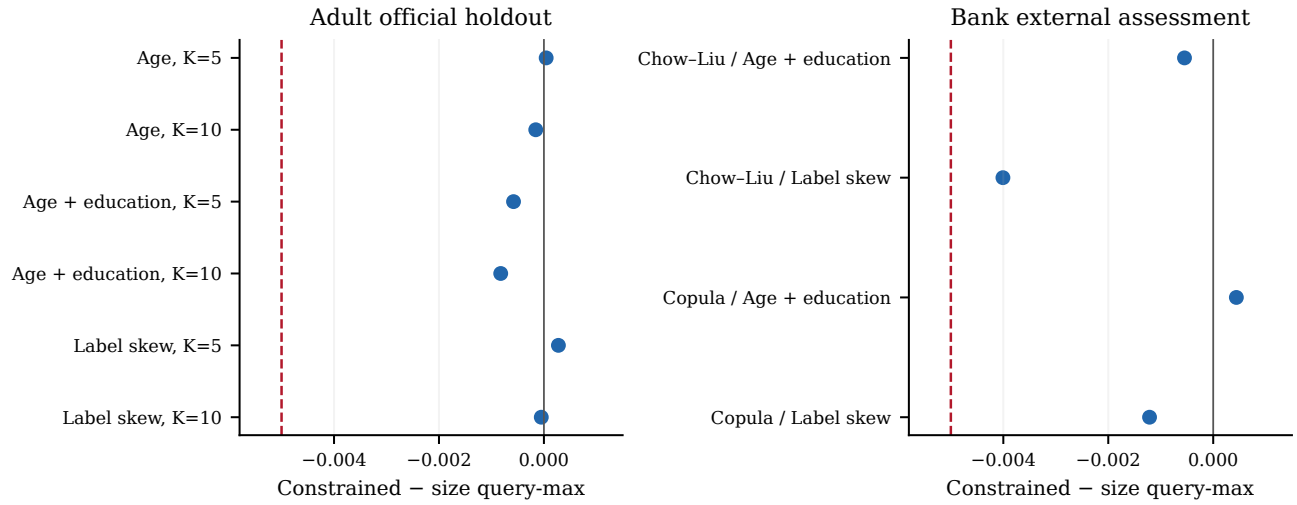


Fig. 2. Frozen comparisons. Left: six Adult partition/K slice means over two fitting seeds. Right: four Bank family/partition slice means over three fitting seeds. Each cell averages five draws. Negative values favour constrained weighting; the dashed margin is -0.005. These are descriptive paired differences with reused people, not independent-cohort estimates.

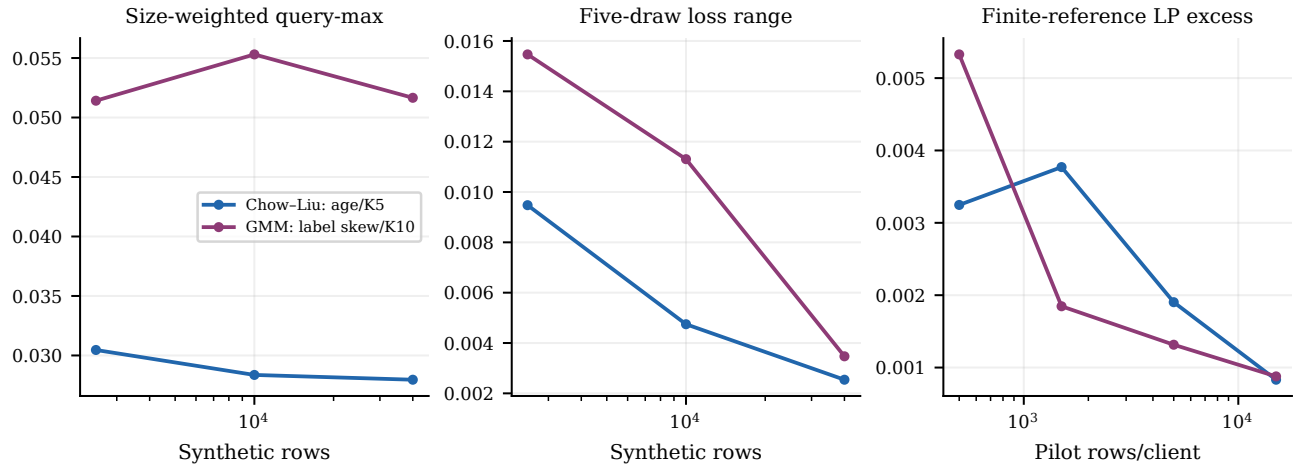


Fig. 3. Development sensitivity on six reused checkpoints. Left: size-weighted primary loss versus generated row count. Middle: within-cell five-draw loss range, averaged over three seeds. Right: same-calibration-target excess relative to an independent finite reference-matrix LP versus pilot rows/client, averaged over pilot repetitions and fitting seeds. Lines connect tested settings; they are not fitted asymptotic laws.

TRTR values 0.101987 and 0.904646. Constrained synthetic training yields Brier 0.129254 and AUROC 0.857161. These small changes do not reverse the main fidelity conclusion. The real and synthetic training budgets differ; no matched-size learning curve was run. Consequently, the comparison measures utility at the registered budgets, not an intrinsic synthetic-versus-real penalty at equal sample size.

Client and subgroup metrics retain their support counts. AUROC is not computed as a supported group claim without at least 20 positives and 20 negatives. Missing or rare subgroup estimates are not filled with global values. Group metrics reuse the same official people, and unsupported group outcomes do not disappear from the evidence tables.

Experiment 7 scores both income labels under the projected joint model for the same six development reference cells. The exact Chow–Liu route has size-weighted mean Brier 0.128481 versus 0.137953 for its synthetic-trained logistic comparator. The approximate GMM score route has 0.126214 versus 0.116265, respectively. Both have finite score coverage for every observed assessment row; this is not universal off-support coverage. The comparison distinguishes a normalized exact model mass from an approximation and from a separately learned predictor. It establishes no general superiority of the joint route.

F. Queries and Mechanisms: Experiments 8–10

Experiment 8 examines every saved query, including 10,500 development answers in the targeted 175-query audit and the frozen official outputs. The conjunction of high-school graduation and high income is the dominant primary error in 33 of the 60 size-weighted confirmation draws. The 23 additional audit queries were viewed during development; their later

TABLE VII
FROZEN SAME-PERSON ADULT UTILITY AT REGISTERED UNEQUAL TRAINING BUDGETS; MEANS OVER 12 CELLS.

Learner	Data/rule	Brier ↓	AUROC ↑	Log loss ↓
Logistic	TSTR size	0.129818	0.856167	0.402970
Logistic	TSTR constrained	0.129254	0.857161	0.401321
Logistic	TSTR uniform	0.132903	0.857331	0.410462
Logistic	TRTR privileged	0.101987	0.904646	0.319003
Extra-trees	TSTR size	0.133218	0.846439	0.413030
Extra-trees	TSTR constrained	0.132705	0.846115	0.411171
Extra-trees	TSTR uniform	0.138131	0.848441	0.423750
Extra-trees	TRTR privileged	0.105741	0.899013	0.329882

TRTR train/tune/calibration: 21,148/4,880/3,254 real records. TSTR: 10,000/2,000/2,000 generated records. Scores use alpha-standardized official reference weights and source-specific calibration. No matched-size intrinsic utility effect is inferred. AUPRC and group-support results remain in the editable evidence tables.

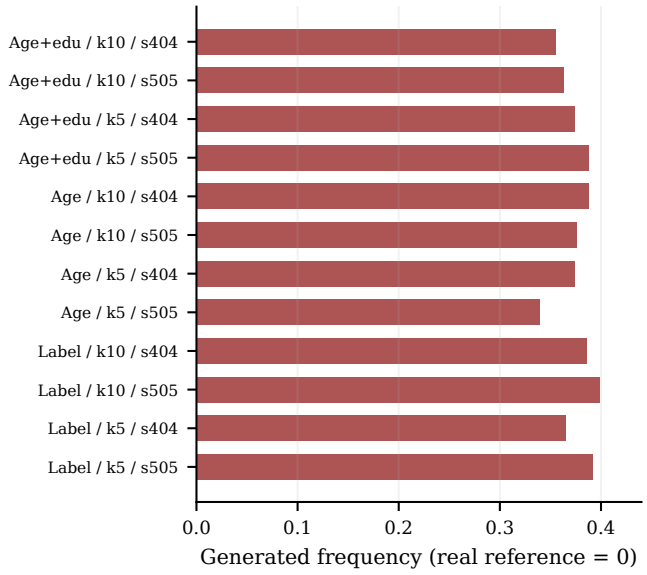


Fig. 4. Female-conditional dependency failure in all 12 frozen Adult cells. Each bar is the generated frequency of `relationship=Husband`; the corresponding alpha-standardized empirical real frequency is zero in every cell. All accepted rows satisfy the female filter.

values are registered secondary descriptions, not independent corroboration of a post-hoc query search.

Experiment 9A applies projection to exactly the same raw generated rows at 12 checkpoints. Across the registered slices, primary maxima fall from approximately 0.268–0.328 to 0.028–0.033 for Chow–Liu and from 0.693–0.708 to 0.049–0.055 for GMM. The maximum includes the education-code relation query. This isolates a structural transformation effect and explains why historical unprojected results cannot be pooled with the corrected core. It does not prove that all other queries improve.

Experiment 9B fixes six copula checkpoints and pilots while varying released summary access: 143 one-way, 151 one/two-way, or all 152 primary queries. Mean differences from size are +0.000208 for one-way and +0.001502 for the other two. The 151- and 152-query solutions coincide in all six cells because the added structural query is already zero on the compared pilots and targets. This null result is specific to that redundant constraint; it does not show that arbitrary higher-order information is useless.

Experiment 10 fixes $K = 5$, exactly 1,000 training and 200 tuning rows per client, and $\alpha_i = 1/5$, using seeds 711/722/733 at assignment severities 0.25 and 0.85. This is a separate equal-target mechanism experiment, not the main natural-size estimand. The range of client mean training ages increases from about 5.1–5.8 to 21.3–21.8 years. Mean primary loss is 0.034804 size versus 0.033317 constrained at low severity, and 0.034918 versus 0.033541 at high severity. Per-seed severity effects change sign. Greater achieved age separation does not produce a stable practically large constrained gain in this batch. Fixing quantity does not isolate every conditional-distribution or assignment effect.

G. Transfer, Privacy, and Resources: Experiments 11 and 12

Experiment 11 checks 48 finite empirical conditional-bound rows and 5,112 student/direct query triangles, supported by 6,816 teacher query answers. Their algebra is covered by Proposition 9; observed checks do not turn the identities into population confidence bounds. Small condition mass can render the bound uninformative, motivating explicit abstention without establishing a real-support guarantee.

Experiment 12 records the nonprivate main release and two separate privacy-related references. The public-bin histogram uses explicit capital-gain/loss zero atoms. Under fixed public membership, sizes, and bins, its 15-column replacement sensitivity is 30, and fresh Laplace count noise of scale $30/\varepsilon$ supplies its stated pure-DP mechanism. One overlapping triplet of $\varepsilon = 1, 4, 8$ fits composes to 13; additional overlapping fits require further accounting. The matched nonprivate histogram still has approximately 0.33 primary error, indicating substantial model-family misspecification. Attack AUC near 0.5 is an attack diagnostic, not a privacy certificate.

Four nonprivate Private-PGM fits use the same released unnoised summaries. Their mean primary difference from size is +0.002615, with one favourable and three adverse cells. Four adapted author-code FLAIM fits have mean difference +0.012894 from a *nonprivate* size reference, again mixed by cell. Each reports $(\varepsilon, \delta) = (4, 10^{-6})$. Subject to valid per-fit assumptions, basic composition of the four released fits is $(16, 4 \times 10^{-6})$, not pure $\varepsilon = 16$ privacy. FLAIM's interactive raw-client contract and private marginals differ from our model/summary contract, so these numbers are not an

equal-access or equal-privacy ranking. The main pipeline is not protected by either isolated mechanism.

Matched application serialization uses pickle protocol 5 and gzip level 6 for both raw-table and model-plus-metadata payloads in three copula cells. Ratios are 2.866 for age-plus-education/$K = 5$, 1.984 for age-plus-education/$K = 10$, and 1.915 for label skew/$K = 10$. Additional aggregate summary traffic is 145,317, 157,267, and 160,769 bytes, respectively. There is no measured compression benefit. Unix-socket application bytes exclude WAN, TLS, authentication, and retries.

H. Closure of the Twelve Packages

Table VIII gives the completed experimental scope. Eleven packages close as bounded findings, and subset composition closes as negative or inapplicable to new expressivity. None implies every early ambition was implemented. In particular, a learned cross-condition service, a feature-adaptive shared/private generator, and an end-to-end private synthesis system remain outside the retained empirical contribution.

VII. Discussion and Limitations

A. What the Negative Result Establishes

The tested constrained calibration rule did not establish the registered practical improvement over tuned, schema-corrected size weighting at the declared target. This statement is supported at three distinct evidence stages: corrected Adult development, frozen Adult scoring, and a prospectively specified Bank replication. It does not mean that every paired difference is positive or exactly zero. It does not prove statistical equivalence, rule out smaller improvements, or show that no other weighting method can help.

The historical copula null and age-only Chow–Liu/GMM gains are useful motivations, not independent confirmations. Later structural audits identified a workload blind spot and a large education-code defect. Corrected comparisons retuned projected generators and matched workloads and checkpoints. Some apparent gains also required mass shifts beyond the representation budget. Those differences in support, tuning, and access prevent a simple narrative that one historical family proves weighting successful while another disproves it.

The geometry explains a genuine limitation: reweighting moves an answer vector inside an existing convex hull. It can correct errors only when the target lies in a helpful direction within that set and the correction respects the feasible representation region. It cannot learn a missing within-client dependency. Better local fitting may reduce correction headroom, while finite-pilot error can make a flat objective yield unstable weights. The observations are consistent with these mechanisms but do not identify a universal causal decomposition.

B. Why Conditional and Utility Failures Matter

The female-conditional failure demonstrates that a modest global primary query maximum can coexist with a large conditional error. The global workload does not comprehensively test every interaction. Similarly, the observed advantage of privileged real training at its larger registered budget does not prove a universal disadvantage of synthetic training; it demonstrates a limit of this realized pipeline for the specified predictive task. Matching source information, training size, tuning, calibration, and access would define a different comparison.

Independent empirical measurements remain necessary even when theoretical identities hold. Bayes routing produces the conditional of the fitted generator, not automatically the real conditional. A student can agree with a flawed teacher. A schema projection can enforce one dependency while leaving another incorrect. A DP mechanism can protect records while having poor utility; a high-utility model can remain nonprivate. These are distinct axes of evaluation.

C. Threats to Validity

The work is a single-host federation simulation with public benchmark records. Adult partitions are artificial institutions, not observed organizational populations. Development splits overlap across seeds, confirmation reuses official people, and Bank can contain repeated contacts. No independent-person or temporal generalization claim is made.

The primary and secondary workloads are finite and partly informed by development diagnostics. Five draws quantify conditional generation variation and are not five independent model fits or cohorts. The selected copula's nominal ordering and approximate score are modelling restrictions. A six-setting local grid is a declared finite search, not exhaustive optimal tuning. The fixed practical margin and representation budget are operational choices rather than theorem-derived universal constants.

Four-cell published references are limited and access-mismatched where stated. HeteroFedSyn was not reproduced numerically. The release ledger contains unnoised models, statistics, and metadata. Main-pipeline privacy, secure aggregation, transport security, and communication compression are unestablished. No proof in this manuscript upgrades those empirical facts. The theoretical fallback is sufficient and may be vacuous; it is not presented as an experimentally validated new method.

D. Surviving Contribution and Falsifiability

The contribution is the controlled comparison and the explicit connection between target preservation, finite statistical access, and downstream limitations. It offers a reproducible benchmark and a disciplined size-weight fallback rather than an algorithmic superiority claim. A stronger local family, independently fixed interaction workload, or permitted information source could change the observed value of weighting. A new genuinely matched private comparison could change the relative utility ranking. Such evidence would require a separately frozen study; it must not be obtained by retuning on the already scored Adult test.

The theoretical developments specify what stronger evidence would have to control: target-summary error, pilot approximation, finite-draw allocation, conditional mass, loss-class discrepancy, and the complete released transcript. We

TABLE VIII
TWELVE COMPLETED WORK PACKAGES AND THEIR EXPLICIT EXCLUSIONS. EXPERIMENT 3 IS A NEGATIVE/INAPPLICABLE EXPRESSIVITY RESULT; THE OTHER ELEVEN ARE BOUNDED FINDINGS.

No.	Question	Completed scope	Finding	Exclusion
1	Fixed-target weighting	54 Adult core; 12 frozen cells; 12 Bank cells	No registered practical-gain slice	Other targets and model families
2	Finite-pilot optimality	57 source LPs; six repeated-pilot cells	Finite optima; no fresh practical gain	Population/nonlinear optimum
3	Subset composition	Algebra and saved effective-weight check	Mixtures collapse; negative closure	New within-subset estimator
4	Synthetic size	Six checkpoints; three sizes; five draws	Draw range shrinks; error persists	Universal asymptotic floor
5	Conditional sampling	Six student cells; 12 frozen service cells	Common filters pass; rare requests abstain	Arbitrary-condition fidelity
6	Downstream utility	Logistic/trees; same-person scoring	TRTR better at larger recorded budget	Matched-size superiority/inferiority
7	Joint-implied prediction	Six projected development cells	Exact Chow–Liu; approximate GMM	Universal support/exact continuous score
8	Query diagnostics	175 queried diagnostics and frozen answers	Localized global and conditional failures	All-query/population TV accuracy
9	Controlled ablations	12 projection; six summary-access cells	Repair matters; redundant-query null	Unrun component factorials
10	Heterogeneity mechanism	Six fixed-quantity, equal-target cells	Skew achieved; no practical gain	Universal causal decomposition
11	Conditional transfer	48 bound rows; 5,112 triangles	Finite algebra; low-support amplification	Population confirmation from identities
12	Privacy/resources	Histogram; four PGM/four FLAIM; three payload cells	Main nonprivate; payload ratios above one	End-to-end DP, secure/WAN deployment

establish these implications and their boundaries here. We do not claim priority for elementary convexity, concentration, Bayes identities, or privacy composition.

VIII. CONCLUSION

Within the registered non-neural, fixed-target study, constrained weighting did not establish a practical fidelity improvement over tuned projected size weighting. Frozen Adult and external Bank comparisons preserve that conclusion at the declared 0.005 margin. The accompanying proofs delimit mixture expressivity, uncertainty, representation shifts, conditional transfer, and downstream implications; the empirical audits expose meaningful support, dependency, utility, and release-cost limitations. The resulting thesis is a bounded comparative negative result with a reproducible statistical framework. It supports retaining the specified size-weighted procedure for this benchmark while leaving broader superiority and privacy claims unestablished.

APPENDIX A
UNCERTAINTY, VALIDATION, AND ADAPTIVITY

A. Synthetic Pilot Error

Proposition 11 (Simultaneous pilot accuracy)**.** *Conditional on fitted models and a fixed workload, suppose each column of $\widehat{H}$ averages p_i independent draws from Q_i. With probability at least $1-\gamma_H$,*

$$\left\|\widehat{H}_{\cdot i}-H_{\cdot i}\right\|_\infty \le \sqrt{\frac{\log(2KL/\gamma_H)}{2p_i}} \quad \textit{for every } i. \tag{27}$$

The maximum of these client-specific radii is a valid r_H in Theorem 5.

Proof. For each client/query pair Hoeffding's bound is $2\exp(-2p_iu_i^2)$. Set each to $\gamma_H/(KL)$ and sum over KL events. Independence between different queries is unnecessary. Fresh within-client sampling is sufficient; data-derived models and queries can be conditioned upon if pilot draws are independent of their construction. □

Selecting weights adaptively from a fixed matrix does not require a union bound over infinitely many weights because the column bound is uniform over the entire simplex. Selecting queries after inspecting the same pilot, selecting among unaccounted repeated pilots, or reusing teacher samples for both fitting and evaluation is a different operation. It requires a predeclared finite union, independent data, or a valid uniform-complexity argument.

B. Calibration Versus a Real Empirical Reference

The desired b_R is not known to a release-only server. A theorem needs a reason that permitted $\widehat{b}$ is close to it. The following is one sufficient sampling model, not an assumption established by a finite benchmark partition.

Proposition 12 (A sufficient target-summary radius)**.** *Fix α and $\mathcal{Q}$. For each i, suppose calibration and reference observations are independent, bounded-query evaluations of sizes $c_i, t_i > 0$, drawn from the same client law; observations are independent within and across these samples. Let b_{cal} and b_R use the same α. Then with probability at least $1-\gamma_b$,*

$$\|b_{\mathrm{cal}}-b_R\|_\infty \le r_{\mathrm{data}} = \sqrt{\frac{\log(2L/\gamma_b)}{2}\sum_i \alpha_i^2\left(\frac{1}{c_i}+\frac{1}{t_i}\right)}. \tag{28}$$

If $\widehat{b}=b_{\mathrm{cal}}+e$ and $\|e\|_\infty \le r_{\mathrm{noise}}$, then $r_b = r_{\mathrm{data}}+r_{\mathrm{noise}}$ is valid. For exact released summaries and a calibration target itself, $r_b=0$; this does not make that target identical to a held-out reference.

TABLE IX
THEORY COVERAGE AND EVIDENTIAL BOUNDARY. REFERENCES IDENTIFY RESULTS PROVED IN THIS MANUSCRIPT.

Original object or guarantee	Present treatment	Boundary retained
Non-IID source/target identity	Eq. (1); Sec. II	Population interpretation does not establish finite-sample independence.
Joint release and aggregation	Secs. II–III	Model-only, summary, and privileged access are separate.
Oracle, hull, identification	Eq. (13); Prop. 3; Counterexample 4	No universal oracle recovery from unrestricted model releases.
Representation and error terms	Props. 1–2; Eq. (48)	Sharp finite-query bound; population terms unmeasured.
Regularized weighting and stability	Theorem 5; Prop. 6	Stable selection does not identify true target weights.
Pilot and target uncertainty	Props. 11–12; Cor. 13	Requires declared sampling/adaptivity assumptions.
Synthetic sample-size error	Theorem 7	Covers fixed allocation; cannot remove fitted-model error.
Uncertainty-aware size fallback	Theorem 8	Sufficient new analysis; not evaluated or used for frozen selection.
Subset compositions	Prop. 3	No expanded family from regrouping mixtures.
Projection and exact scores	Prop. 14	Exact only with exact masses/integrated preimages.
Conditional routing and transfer	Eq. (24); Prop. 9; Cor. 17	Positive mass and relevant joint queries are required.
Student filtering and support	Props. 15–16	Acceptance, internal consistency, and real fidelity differ.
Joint predictor and utility	Eq. (26); Props. 18–19	Primary synthesis workload does not control all losses.
Finite geometric/transport pools	Props. 20–22	Support, smoothing, cost, and search gaps remain explicit.
Privacy/composition	Eq. (46); Prop. 23	Conditional isolated mechanisms; main transcript nonprivate.
Communication and compression	Appendix E	Measured payloads exceed matched raw tables; no WAN claim.

Proof. For each query, the calibration-minus-reference mean is a sum of independent mean-zero bounded variables. Their interval lengths are α_i/c_i repeated c_i times and α_i/t_i repeated t_i times. The sum of squared lengths is $V = \sum_i \alpha_i^2(1/c_i + 1/t_i)$. Hoeffding gives $2\exp(-2r^2/V)$; a union bound yields (28). Add the release error by the triangle inequality. □

This theorem is conditional on a training-fitted design when such conditioning leaves the subsequent samples independent with fixed queries and masses. It cannot simply be asserted for group-constrained sampling without replacement, adaptive partition selection, repeatedly viewed assessment queries, or repeated contacts. The appropriate alternative is a finite-population design bound or an explicitly empirical result. The present paper uses the empirical interpretation for the campaign.

Under fixed public sizes and replace-one adjacency, a vector of L bounded client query means has ℓ_1 sensitivity at most L/c_i. Independent Laplace coordinates of scale $s_i = L/(c_i\varepsilon_i)$ have, by a union bound,

$$\max_{i,\ell} |\xi_{i\ell}|/s_i \le \log(KL/\gamma_\xi) \tag{29}$$

with probability $1-\gamma_\xi$. If $e = \sum_i \alpha_i \xi_i$, then $r_{\text{noise}} = \sum_i \alpha_i s_i \log(KL/\gamma_\xi)$ suffices. This is an illustrative private-summary radius; the actual main summaries are unnoised. Combining (28), (29), and Proposition 11 uses the sum of failure budgets. It supplies no protection for an independently nonprivate model release.

C. A Finite-Reference Validation Oracle

Corollary 13 (Validation-to-reference comparison)**.** *For the same model-answer map H, suppose $\|b_{\text{val}} - b_R\|_\infty \le r_V$. If w_V minimizes $\|Hw - b_{\text{val}}\|_\infty$ over $\mathcal{W}$ to tolerance ζ, then*

$$L_R(w_V) \le \inf_{w\in\mathcal{W}} L_R(w) + 2r_V + \zeta. \tag{30}$$

If H is replaced by a pilot matrix, add $2r_H$.

Proof. The reverse triangle inequality bounds the difference of the two objectives by r_V for every w. Apply this at w_V and at a comparator minimizing L_R, with the optimization inequality between them. The pilot statement follows from Theorem 5. □

A validation optimum is not a test optimum, so a method can have negative test excess relative to it without contradiction. If the solver lacks an optimality certificate, the reference must be called best-found. The LP result does not certify a nonlinear operator's optimum. Moreover, a theorem about minimizing a surrogate is not an assertion that the finite-draw mean actually used for global family selection equals that surrogate.

APPENDIX B
PROJECTION, SUPPORT, AND CONDITIONAL STUDENTS

Proposition 14 (Projected probability and information loss)**.** *Let $T : \mathcal{Z} \to \mathcal{Z}$ be deterministic. On a finite or countable support,*

$$(T_\# Q)(z) = \sum_{u:T(u)=z} Q(u). \tag{31}$$

This is normalized and satisfies $\mathbb{E}_{T_\# Q} q = \mathbb{E}_Q(q \circ T)$. For arbitrary measures A, B, $\mathrm{TV}(T_\# A, T_\# B) \le \mathrm{TV}(A, B)$.

Proof. The preimages partition the original support, so their masses sum to one. The expectation identity follows by grouping terms by image. For any event E, $(T_\# A)(E) - (T_\# B)(E) = A(T^{-1}E) - B(T^{-1}E)$, bounded by the original TV. Take the supremum. □

If real records are fixed by T, the comparison obeys $d_{\mathcal{Q}}(T_\# Q, R) = d_{\{q\circ T : q\in\mathcal{Q}\}}(Q, R)$. It need not be at most $d_{\mathcal{Q}}(Q, R)$ unless the workload is suitably closed under composition with T. Thus exact schema repair has a clear support guarantee but no automatic improvement for every pre-existing finite-query metric. For continuous-to-discrete rounding, (31) is replaced by integration over preimage cells. Evaluating a continuous density at rounded values omits those integrals and clipping atoms.

For the education projection, exact preimage summation costs a factor proportional to the 17 permitted education-number values per requested row. General projection maps may have exponentially many preimages; normalization of the

original model does not make marginalization computationally trivial. The recorded discrete audit supports exactness for the implemented map/model, not arbitrary projections.

Proposition 15 (Support-pilot statement and rejection cost)**.** *For a fixed condition C and p independent IID proposals from a model Q, let $\widehat{c}$ be the matching fraction and $c = Q(C)$. With probability $1-\gamma$,*

$$c \geq \widehat{c} - \sqrt{\log(1/\gamma)/(2p)}. \tag{32}$$

For a largest-remainder pilot from mixture Q_w, subtract an additional $\rho_p(w)$ to obtain a bound on $Q_w(C)$. For J predeclared conditions, replace γ by γ/J. Under IID rejection with $c > 0$, the expected proposals for a accepted rows is a/c, and the probability of failing a cap B is $\mathbb{P}\{\mathrm{Binomial}(B, c) < a\}$.

Proof. Apply the one-sided Hoeffding inequality to indicators. With fixed client counts, indicators are independent but may have different means; their average expectation is $Q_{\bar{w}}(C)$, within $\rho_p(w)$ of $Q_w(C)$. For IID proposals, waiting times between accepts are geometric with mean $1/c$; their sum has mean a/c, and accepted counts before the cap are binomial. □

The frozen threshold of 300/30,000 is an operational rule, not the lower-confidence rule above. These statements concern synthetic-model support, not real support or conditional accuracy. The binomial and geometric expressions require IID proposals. A batch service using fixed client quotas and truncation must not be described as exactly IID rejection without verifying its proposal and ordering law. Its actual acceptance counts and costs are the empirical evidence in this study.

Proposition 16 (Student filtering and transfer)**.** *Let M be a raw per-condition student distribution with $a = M(C) > 0$, and let $M_C = M(\cdot \mid C)$. Then $\mathrm{TV}(M, M_C) = 1 - a$. For any fixed conditional workload and teacher conditional T_C,*

$$d_{\mathcal{Q}_C}(M_C, R_C) \leq d_{\mathcal{Q}_C}(M_C, T_C) + d_{\mathcal{Q}_C}(T_C, R_C). \tag{33}$$

An average error over conditions does not bound a worst-case error without further support or complexity restrictions.

Proof. Write $M = aM_C + (1-a)M_{C^c}$, with disjoint conditional supports; the event C^c attains TV $1-a$. The transfer inequality is the querywise triangle inequality followed by a maximum. For the last statement, place an error of one on a condition of evaluation probability η and zero elsewhere. The average is η while the maximum is one, for any positive η. □

The student can satisfy C perfectly after filtering and still be arbitrarily wrong within C. A separately fitted model for each of four requests does not establish generalization to an unseen condition. Consistency with a teacher is a useful internal metric, but a teacher can itself be wrong.

For an explicitly declared distribution ν over requests, a conceptual distillation objective is

$$\inf_{\phi} \ \mathbb{E}_{C\sim\nu}\left[d_{\mathcal{Q}_C}\left(M_\phi(\cdot \mid C), Q_w^X(\cdot \mid C)\right)\right]. \tag{34}$$

It separates approximation of the teacher, finite teacher-training samples, and optimization error. The implemented student fits are separate per registered condition, not a learned map from unseen requests to distributions or an optimizer certified for this expression. Define conditional consistency by the displayed expectation and predictive consistency by $\mathbb{E}_{Q_w^X}(g_S - \eta_{Q_w})^2$. Real-reference fidelity must be measured separately for both. This distinction preserves the original three-object framework: joint generator, conditional feature model, and income-probability predictor.

Corollary 17 (Abstract TV conditional implication)**.** *For arbitrary measures A, B, suppose $\mathrm{TV}(A, B) \leq e$ and $A(C) \geq \tau > e$. Then $B(C) > 0$ and*

$$\mathrm{TV}(A(\cdot \mid C), B(\cdot \mid C)) \leq \min\{1, 2e/\tau\}. \tag{35}$$

A separate conditional student with TV error a_C from $B(\cdot \mid C)$ adds a_C to this upper bound, capped at one.

Proof. $B(C) \geq A(C) - e > 0$. Apply the proof of Proposition 9 to every indicator of an event inside C: both joint-event and mass discrepancies are at most e. Take the event supremum and then the triangle inequality for the student. □

This preserves the earlier theoretical population implication without pretending its premise was measured. A small finite-workload error cannot be substituted for e in this corollary. Rare conditioning necessarily amplifies error: altering all outcomes within a region of mass τ can change an unconditional event by only τ while reversing its conditional probability by one.

Appendix C
Prediction and Bounded-Loss Utility

Let $\mathcal{G}$ be a class of predictors and let $\ell(g(x), y) \in [0, 1]$. For a measure A define $R_A(g) = \mathbb{E}_A \ell(g(X), Y)$ and the loss-class discrepancy

$$d_{\ell,\mathcal{G}}(A, B) = \sup_{g\in\mathcal{G}} |R_A(g) - R_B(g)|. \tag{36}$$

This is a different query class from the 152 synthesis queries. If all of its loss functions were covered by a suitable workload bound, risk transfer would follow; the implemented workload does not supply that premise.

Proposition 18 (Risk transfer and approximate empirical learning)**.** *Suppose $d_{\ell,\mathcal{G}}(R_\alpha, Q_w) \leq e_\ell$ and a synthetic sample satisfies $\sup_{g\in\mathcal{G}} |\widehat{R}_{D_S}(g) - R_{Q_w}(g)| \leq \xi_m$. If g_S is an approximate synthetic empirical-risk minimizer with tolerance ζ, then*

$$R_{R_\alpha}(g_S) - \inf_{g\in\mathcal{G}} R_{R_\alpha}(g) \leq 2e_\ell + 2\xi_m + \zeta. \tag{37}$$

For arbitrary measures A, B, $d_{\ell,\mathcal{G}}(A, B) \leq \mathrm{TV}(A, B)$. For finite fixed $|\mathcal{G}| = M$ and IID synthetic draws, one sufficient radius is $\xi_m = \sqrt{\log(2M/\gamma)/(2m)}$ with probability $1-\gamma$.

Proof. For any comparator g, move from real risk of g_S to model risk, then empirical synthetic risk, use approximate empirical optimality, and reverse the two comparisons for g. This incurs two copies of e_ℓ and ξ_m, plus ζ; take the infimum. The TV implication follows by integrating the layer-cake indicators of a $[0, 1]$-valued loss. The finite-class statement

is Hoeffding plus a union bound. Stratified generation adds the allocation correction as in Theorem 7. □

For infinite or adaptively chosen predictor classes a capacity bound or other valid uniform argument replaces the finite-class union. Logistic regression with its fitting and calibration pipeline is not certified by pretending it is the finite hyperparameter grid alone. Clipped binary log loss is bounded by $-\log(10^{-6})$, so the corresponding TV risk bound is scaled by that constant. AUROC is a pairwise, class-conditional metric and is not a direct application of the displayed one-record bounded-loss result.

Proposition 19 (Brier decomposition and teacher consistency)**.** *For binary Y, let $\eta_A(x) = A(Y = 1 \mid X = x)$. For any $g : \mathcal{X} \to [0, 1]$,*

$$R_A(g) - R_A(\eta_A) = \mathbb{E}_{A^X}(g - \eta_A)^2. \tag{38}$$

If $\mathrm{TV}(A, B) \le e$ and $a_g = \mathbb{E}_{B^X}(g - \eta_B)^2$, then

$$\mathbb{E}_{A^X}(g - \eta_A)^2 \le a_g + 2e. \tag{39}$$

Proof. Condition on X: $\mathbb{E}[(g - Y)^2 \mid X] = (g - \eta_A)^2 + \eta_A(1 - \eta_A)$. Subtract the Bayes risk and integrate. For transfer, $R_A(g) \le R_B(g) + e = R_B(\eta_B) + a_g + e \le R_B(\eta_A) + a_g + e \le R_A(\eta_A) + a_g + 2e$. □

Predictive consistency a_g and conditional consistency $d_{\mathcal{Q}_C}(M_C, Q_w^X(\cdot \mid C))$ quantify module agreement. Neither establishes agreement with real records. No theorem here promises synthetic training to outperform real training. Conversely, an unequal-budget empirical disadvantage does not prove it must always underperform.

Appendix D
Finite-Support Operators and Approximation

This appendix completes the theoretical treatment of the nonlinear candidate families while separating it from their finite implementation outcomes. All support sizes, floors, costs, and decision sets must be specified for a numerical certificate.

Proposition 20 (Restriction and smoothing on a finite support)**.** *Let a discrete target model Q assign positive mass to finite candidate support $\mathcal{S}$, and let $Q_{\mathcal{S}} = Q(\cdot \mid \mathcal{S})$. Then*

$$\mathrm{TV}(Q, Q_{\mathcal{S}}) = Q(\mathcal{S}^c). \tag{40}$$

If $\widehat{Q}_{\mathcal{S}}$ has normalized mass-vector error at most η in ℓ_1, and smoothing forms $Q' = (1 - \beta)\widehat{Q}_{\mathcal{S}} + \beta U$, then

$$d_{\mathcal{Q}}(Q, Q') \le \min\{1, Q(\mathcal{S}^c) + \eta/2 + \beta\}. \tag{41}$$

Applying the same deterministic projection to both sides does not enlarge the TV version of this bound.

Proof. Restriction is the filtering identity of Proposition 16. On a discrete support TV is half the ℓ_1 mass difference. Mixing in βU changes TV by at most β. Sum these quantities, use $d_{\mathcal{Q}} \le \mathrm{TV}$, and apply Proposition 14. □

Candidate support can omit most of a continuous model's mass; conditioning a continuous model on finitely many exact points is not this construction. A finite empirical pool is then a separate operator, with query or transport approximation assessed directly. Even in the discrete case, the missing-support term may be unknown, so this formula is not automatically an available certificate.

Proposition 21 (Geometric-score perturbation)**.** *On common finite support, let positive unnormalized masses be $g(z) = \exp h(z)$ and $g'(z) = \exp h'(z)$, with $\max_z |h'(z) - h(z)| \le \eta$. Normalize them to G, G'. Then*

$$\begin{aligned} e^{-2\eta} \le G'(z)/G(z) \le e^{2\eta}, \\ d_{\mathcal{Q}}(G, G') \le \min\{1, (e^{2\eta} - 1)/2\}. \end{aligned} \tag{42}$$

For a fixed geometric weight vector, client log-score perturbations η_i give $\eta \le \sum_i w_i \eta_i$.

Proof. The pointwise unnormalized ratio lies in $[e^{-\eta}, e^{\eta}]$, as does the ratio of normalizing constants. Dividing gives the first bound. Sum $\frac{1}{2}G(z)|G'(z)/G(z) - 1|$ and bound the absolute ratio deviation by $e^{2\eta} - 1$. The weighted score statement is the triangle inequality. □

A floor replacing an exact zero has no finite log-error relative to the zero. Thus smoothing can change the operator substantially; it is not justified by this proposition without a positive-score premise. Identical-support scoring and exact normalization are prerequisites even before a real-data comparison.

Proposition 22 (Finite entropic transport objective boundary)**.** *On finite probability supports with at most B points each, define*

$$T_\varepsilon(a, b) = \min_{\pi \in \Pi(a,b)} \langle C, \pi \rangle + \varepsilon \,\mathrm{KL}(\pi \| a \otimes b), \tag{43}$$

and T_0 without the penalty. Then $T_0 \le T_\varepsilon \le T_0 + \varepsilon \log B$. For fixed client weights, if a_ε minimizes $F_\varepsilon(a) = \sum_i w_i T_\varepsilon(a, b_i)$ to tolerance ζ over a common allowed set $\mathcal{A}$, then

$$F_0(a_\varepsilon) - \inf_{a \in \mathcal{A}} F_0(a) \le \varepsilon \log B + \zeta. \tag{44}$$

A cost perturbation bounded entrywise by κ adds 2κ when comparing the perturbed minimizer to the original objective. Restricting $\mathcal{A}$ further adds its approximation gap relative to a larger decision set.

Proof. For a coupling, $\mathrm{KL}(\pi \| a \otimes b)$ is mutual information and lies between zero and $\min\{H(a), H(b)\} \le \log B$. Evaluate the penalized objective at an unpenalized optimal coupling for the upper bound; nonnegativity gives the lower bound. Hence $F_0 \le F_\varepsilon \le F_0 + \varepsilon \log B$ uniformly, which yields (44) by comparison at its minimizer. A unit-mass coupling changes its cost by at most κ, giving two copies through a minimizer comparison. A restricted minimum differs from the larger-set minimum by exactly the stated nonnegative approximation gap. □

This statement uses the displayed KL-to-product regularization. Other entropy conventions may differ by terms depending on the optimized marginal and must be converted before applying it. A finite search over weights incurs its own search gap; a global continuous barycenter guarantee does not follow. Objective closeness also does not imply

close probability measures without additional identification or curvature assumptions.

For any coupling π of A, B and a query that is L_q-Lipschitz with respect to cost metric d,

$$|\mathbb{E}_A q - \mathbb{E}_B q| \le L_q \mathbb{E}_\pi d(Z, Z'). \tag{45}$$

This follows by integrating $|q(z) - q(z')| \le L_q d(z, z')$. A threshold indicator need not have a small Lipschitz constant in the chosen mixed-data metric. Thus low entropic transport objective is not automatically a guarantee for the actual finite rectangle/equality workload.

APPENDIX E
PRIVACY, COMMUNICATION, AND UNMEASURED POPULATION IMPLICATIONS

A. Complete-Transcript Privacy

For a randomized release $\mathcal{M}$ and a specified neighboring relation $D \sim D'$, (ε, δ)-DP means

$$\mathbb{P}\{\mathcal{M}(D) \in E\} \le e^{\varepsilon} \mathbb{P}\{\mathcal{M}(D') \in E\} + \delta \tag{46}$$

for every measurable E. Standard postprocessing and composition properties are used in their stated scope [13]. The relevant release is the entire transcript, including models, summaries, data-derived schema, tuning decisions, counts, selection, and further real-data queries. Sending a fitted model instead of rows is not a proof of (46).

For d released one-way histogram count vectors on fixed public bins, replacing a row changes at most two count entries per column. Therefore ℓ_1 sensitivity is at most $2d$. Independent Laplace count noise of scale $2d/\varepsilon$ has likelihood ratio at most e^{ε}: the ratio is bounded by the exponential of the count-vector difference divided by the noise scale, using the triangle inequality. This proves the isolated histogram mechanism for fixed membership and public sizes. Clipping, normalizing, projecting, and sampling from its noisy output use no new real records and are postprocessing.

Similarly, the client mean-vector sensitivity used in (29) is L/c_i, because each of L means changes by at most $1/c_i$. This derivation does not assume the queries are statistically independent. It does require that their definitions and denominator are fixed under the neighboring relation.

Proposition 23 (Scope of composed release guarantees)**.** *Suppose each full client mechanism is $(\varepsilon_i, \delta_i)$-DP, clients use independent local randomization, membership is fixed and disjoint, and record replacement affects only one client. Joint release is $(\max_i \varepsilon_i, \max_i \delta_i)$-DP. Subsequent server operations using only that release and independent randomness preserve the guarantee. If multiple mechanisms access overlapping records with the required conditional DP guarantees, basic sequential composition instead sums their ε and δ budgets. These conclusions are conditional on valid guarantees for every released mechanism; reusing hidden random noise is not justified by marginal per-fit guarantees alone.*

Proof. Under fixed disjoint membership, a neighboring record replacement changes only one client output law. Condition on the unaffected outputs and apply that client's DP inequality; integrate and take the largest budgets. Server postprocessing replaces an event by a randomized event probability in $[0, 1]$ and integrates the same inequality. For overlapping accesses, the standard adaptive composition theorem supplies the sum of budgets; disjoint parallel reasoning is inapplicable. The displayed theorem is an application of these established rules, not a new composition result. □

The actual main release includes nonprivate models and unnoised summaries and therefore does not meet the premise. A record changing client under feature-based assignment can affect two client datasets and invalidate the fixed-membership argument. Public alpha and membership are assumptions of an isolated baseline, not protections for data-derived partition metadata. Record-level and client-level privacy protect different neighboring relations.

For one overlapping triplet of the histogram settings, the budget is $1 + 4 + 8 = 13$ under its stated adjacency. Additional released settings or overlapping fitting seeds need further composition; 13 is not a whole-campaign budget. The four adapted FLAIM fits have basic conditional composition $(16, 4 \times 10^{-6})$. Five synthetic draws from one valid private fit do not cost five additional privacy budgets. If both sets of mechanisms were released together under a common valid adjacency, they would also require joint accounting; the nonprivate main transcript still prevents a finite DP claim for the full archive. Empirical membership attacks cannot prove the universal inequality (46).

B. Communication and Compression

Let $B_{\text{models}}, B_{\text{metadata}}, B_{\text{messages}}$ be transmitted application bytes under a stated serialization. Then

$$B_{\text{total}} = B_{\text{models}} + B_{\text{metadata}} + B_{\text{messages}}. \tag{47}$$

For K clients, L scalar summaries, and s bits/scalar, the numerical payload alone contributes $KLs/8$ bytes before encoding and protocol overhead. Fit time, weight optimization, serialization, generation, conditional rejection, and predictor fitting are separate costs. Sharing a model may reduce future per-row generation cost without reducing its initial transmitted size.

If a compressed operator changes every workload answer by at most e_{cmp}, its real-reference model-answer loss is at most the uncompressed loss plus e_{cmp}, by the triangle inequality. This supplies a fidelity accounting term, not a privacy guarantee. Actual compression ratios here are adverse; no compressed-model approximation theorem was empirically instantiated.

C. A Generic Population Decomposition, Without an Empirical Claim

For completeness, the earlier population framework admits the following conditional implication. Let P_i be hypothetical client laws, Q_i fitted models, $e_i = \mathrm{TV}(P_i, Q_i)$, and $D_Q = \max_{i,j} \mathrm{TV}(Q_i, Q_j)$. For any output operator A_w, define $a_A = \mathrm{TV}(\sum_i w_i Q_i, A_w)$. Then

$$\mathrm{TV}\left(\sum_i \alpha_i P_i, A_w\right) \le \sum_i \alpha_i e_i + \frac{D_Q}{2} \|w - \alpha\|_1 + a_A. \tag{48}$$

Insert $\sum_i \alpha_i Q_i$ and $\sum_i w_i Q_i$; mixture convexity bounds the first term. Decompose positive and negative weight differences into two probability mixtures as in Proposition 1; their TV is at most the maximum component diameter, yielding the second term. The operator contributes the third term. If target proportions are estimated, an additional $\frac{1}{2}\left\|\widehat{\alpha}-\alpha\right\|_1$ bound is available. A compressed output contributes its further TV discrepancy.

None of e_i, D_Q as a population certificate, or the left side of (48) was estimated by the reported finite-query experiments. They are not labels for observed dataset distances. The empirical counterpart that is actually relevant to measured tables is Proposition 2. This explicit separation preserves the theoretical implication without turning an unattained premise into a research result.

APPENDIX F
REPRODUCIBILITY, EVIDENCE, AND THEOREM COVERAGE

A. Artifacts and Version Boundaries

The author-ready pack supplies editable results, figures, a claim index, and the experimental closure register. The manuscript source bundle includes these table inputs, the freeze and workload definitions, scripts that regenerate the manuscript tables/plots, and an executable finite-algebra check. It does not duplicate the separately archived full experimental code, raw data, or model checkpoints, and it does not claim a public GitHub release.

The full executable project is archived as BTP_FINAL_EXPERIMENTAL_PACKAGE_20261025_v2.zip. Its reported size is 537,318,039 bytes and SHA-256 is

`b184b0b07d3185ca7bd95322fd7bf432`
`b28293b1041f5cae5cc9153ead25b2bf`

That full archive was not present in the manuscript-authoring environment; its reported 91-test clean extraction is archived execution evidence, not a fresh rerun performed for writing this paper. The supplied documentation/results archive was independently checked against all 3,733 listed file hashes, and the author-ready addendum against all 83 listed hashes. The review archive intentionally excludes original code and checkpoints.

The global selection freeze SHA-256 is

`2420f737b6af732925f9cce3fbb82c8a`
`9a7a30e1988dde5e68900d1404bb3a23`

The canonical confirmation source hash is

`2d7a7598b43865d969dacbb8c91f06ac`
`046b38670e9ef98169c3815dd8b2bb24`

The frozen workload hash is

`b93cd48509ee437600cc0baee01dff2a`
`873553ba3ea1b176d9d6cfbd095d4b0f`

The historical corrected development source hash differs and is retained in the evidence. An archival replay under an old source hash is not a license to modify frozen confirmation scoring.

The earlier independent read-only review reproduced all 180 confirmation synthetic-draw hashes and 31,500 saved synthetic query answers, checked the 36 supported conditional samples, and recomputed global Brier/log-loss summaries from 1,562,976 saved prediction rows. These checks validate consistency of available outputs. They do not replace running the full fitting archive or establish the correctness of unprovided code. The manuscript-generation script separately checks all table denominators and the headline means against its copied CSV inputs.

No new data were fit, no method was selected, and no official test was rescored to prepare this manuscript. Additional graphs are renderings of existing saved tables. Computational checks of mathematical examples are not counted as new empirical work packages or as proofs of theorems.

B. Frozen Implementation Details

Adult numerical public supports are age $[0, 100]$, `fnlwgt` $[0, 2{,}000{,}000]$, education-number $[0, 16]$, capital gain $[0, 100{,}000]$, capital loss $[0, 5{,}000]$, and hours/week $[0, 100]$. The declared vocabularies preserve unknown/missing tokens. The education map contains the 16 recognized education categories. These supports are modelling declarations, not claims about every real-world person's possible attributes.

The main synthetic seed is master seed plus $4200 + r$ for repeat $r = 0, \ldots, 4$. The weighting pilot uses master seed plus $900 + i$ for client i. Client fits use master seed plus $31i+$ grid index; local tuning draws add a further 100. Confirmation utility fit randomness uses master seed plus 12000; synthetic train/tune/calibration substreams add 9000, 10000, and 11000. Conditional support adds 13000 and proposal batches use the recorded condition-specific 14000-based substreams. Public pseudorandom synthesis seeds are distinct from the fresh, nonpublic privacy-noise randomness of isolated DP references.

Within-cell candidate methods share fitted checkpoints, target, workload, and compatible draw seeds. Matched seeds induce useful paired comparisons but do not guarantee identical per-row couplings when allocation weights differ. Global utility uses alpha-standardized row weights. No audit or test endpoint serves as an undisclosed family-selection tie break. Empty positive-mass reference clients, failed fits, unsupported conditional requests, and solver failures have explicit terminal states.

Published-reference source revisions recorded in the full archive are MBI `9429881f0f21` and FLAIM `3e698da752b4`. The HeteroFedSyn source was inspected at `5aec4eac2843`; inspection is not numerical replication. Full revision identifiers are retained in the source-bundle provenance file. Primary literature URLs are supplied in the bibliography; no citation is used to imply a matched comparison that was not run.

C. Coverage of the Original Theory Programme

Table IX maps the earlier programme to the present treatment. Standard identities, proved sufficient conditions, and uninstantiated assumptions remain distinguishable. In particular, an end-to-end DP theorem is not owed for the explicitly nonprivate empirical claim, and a theoretical fallback is not relabelled as the historical frozen method.

D. Reporting and Research Use

Data attribution follows the UCI sources. The tables describe public benchmark records and synthetic derivatives, not a deployment on private institutional populations. Sex, race, relationship, and income outputs are evaluated as dataset fields; the study does not convert observed codes into universal social constraints or causal statements. The public-data experiment does not authorize releasing analogous models of private records without a separate governance and privacy design.

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